

ASSIGNMENT No: 11

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TITLE: Write a Java program to demonstrate the implementation of interfaces.

Problem Statement: Write a Java program to demonstrate the implementation of interfaces.

Learning Objectives: To implement interfaces for achieving abstraction and multiple inheritance of type.

Theory:

An interface specifies a set of abstract methods that implementing classes must define. It provides complete abstraction and supports multiple inheritance of type. Interfaces improve modularity, loose coupling, and software flexibility. They allow different classes to implement common behavior while maintaining independent implementations. Interfaces are widely used in enterprise application development.

Exercise

1. Create an interface Printable and implement it in Student&Employee classes.

```
PrintableDemo.java > Employee > print()
1  interface Printable {
2      void print();
3  }
4
5  class Student implements Printable {
6
7      public void print() {
8          System.out.println(x: "Student Details Displayed");
9      }
10 }
11
12 class Employee implements Printable {
13
14     public void print() {
15         System.out.println(x: "Employee Details Displayed");
16     }
17 }
18
19 public class PrintableDemo {
20     Run | Debug
21     public static void main(String[] args) {
22
23         Student s = new Student();
24         Employee e = new Employee();
25
26         s.print();
27         e.print();
28     }
29 }
```

PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS

[Done] exited with code=1 in 0.196 seconds

[Running] cd "c:\Users\arhaa\OneDrive\Desktop\java\" && javac Student.java Employee.java PrintableDemo.java

Student Details Displayed

Employee Details Displayed

[Done] exited with code=0 in 0.732 seconds

2. Create an interface Switchable with a method turnOn().
Implement the interface in Light and Fan classes to display the device status.

```
J SwitchableDemo.java > ...
1  interface Switchable {
2      void turnOn();
3  }
4
5  class Light implements Switchable {
6
7      public void turnOn() {
8          System.out.println(x: "Light is ON");
9      }
10 }
11
12 class Fan implements Switchable {
13
14     public void turnOn() {
15         System.out.println(x: "Fan is ON");
16     }
17 }
18
19 public class SwitchableDemo {
20     Run | Debug
21     public static void main(String[] args) {
22
23         Light l = new Light();
24         Fan f = new Fan();
25
26         l.turnOn();
27         f.turnOn();
28     }
29 }
```

PROBLEMS 9 OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
PS C:\Users\arhaa\OneDrive\Desktop\java> & 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' -cp 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe'; cd 'C:\Program Files\Java\jdk-11.0.10\bin'; java -cp 'C:\Program Files\Java\jdk-11.0.10\bin\java.exe' SwitchableDemo
Light is ON
Fan is ON
PS C:\Users\arhaa\OneDrive\Desktop\java>
```